

Objective

In this lab test, you will implement two data structures in C++: a singly linked list and a stack using an array. You will work with classes **to represent objects in each case**.

Part 1: Singly Linked List of Objects

Task Description:

Create three classes: **Email**, **Node**, and **TheList**.

- **Email Class:** This class should include two public members: id (int) and password (string).
- **Node Class:** This class contains an Email object and a pointer to the next node (Node* next).
- **ListEmail Class:** This class manages the linked list, including operations to initialize the list, add emails to the end (addBack), delete the first email (deleteFront), and display all emails in the list.

Part 2: Stack Implementation Using an Array of Objects

Task Description:

Modify a previous **Assignment II** where you implemented a stack using a primitive data type. You will now implement a stack using an array of objects. You can use an idea based on the Website case study given.

- Create a _____ class that contains two members:
- Implement a Stack class, which uses an array to manage _____ objects and supports operations such as push, pop, stackTop, isFull, isEmpty and print.

Submission through Elearning

This submission can be considered as backup marks for the Lab Exercise, **especially** for those who did not attend or did not perform well in the previous Lab Exercise. It is not mandatory for students who are confident they can answer Test 2 and have already performed well (*attend and submit*) in the Lab Exercise.

For students who do not have “Assignment II”, you MUST submit this Exercise.

Before 7.1.2025(Tuesday night)